NATIONAL INSTITUTE OF ENGINEERING, MYSURU

Department of CSE – AI & ML

MACHINE LEARNING

BANKNOTE AUTHENTICATION USING LOGISTIC REGRESSION

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1. Problem Statement and Dataset

The objective of this project is to develop and evaluate a Logistic Regression classifier that predicts the class label of a banknote from four numerical measurements extracted from scanned banknote images. The model is assessed on unseen test data and its performance is examined using standard classification metrics and alternative decision thresholds.

Dataset Description

The Banknote Authentication dataset was obtained from the UCI Machine Learning Repository. It contains 1,372 records, four numerical input features, and one binary target variable, Class. The features are Variance, Skewness, Curtosis, and Entropy.

Attribute	Role / description
Variance	Input feature: variance of the wavelet-transformed image
Skewness	Input feature: skewness of the transformed image
Curtosis	Input feature: kurtosis-related measurement (original name: Curtosis)
Entropy	Input feature: entropy of the image
Class	Binary target label: Class 0 or Class 1

Exploratory Data Analysis

The dataset contains 762 records in Class 0 (55.54%) and 610 records in Class 1 (44.46%). The classes are reasonably represented, with a modest difference in counts. The data check identified 24 duplicate rows. The dataset exploration included dimensions, data types, missing-value checks, duplicate checks, descriptive statistics, and class counts.

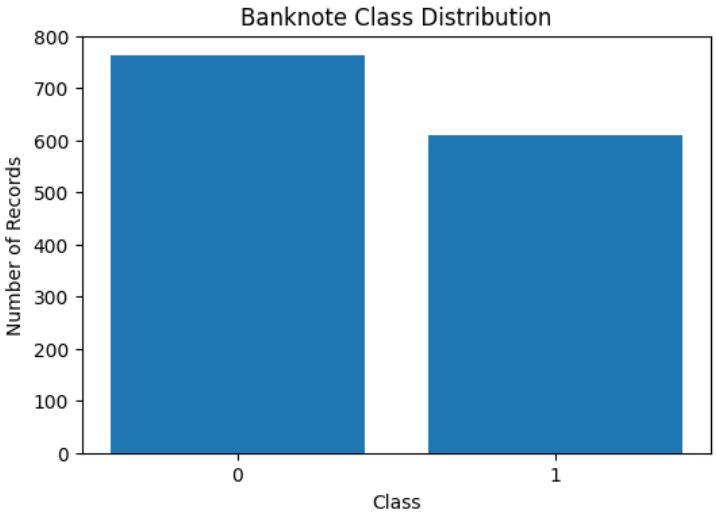


Figure 1. Banknote class distribution

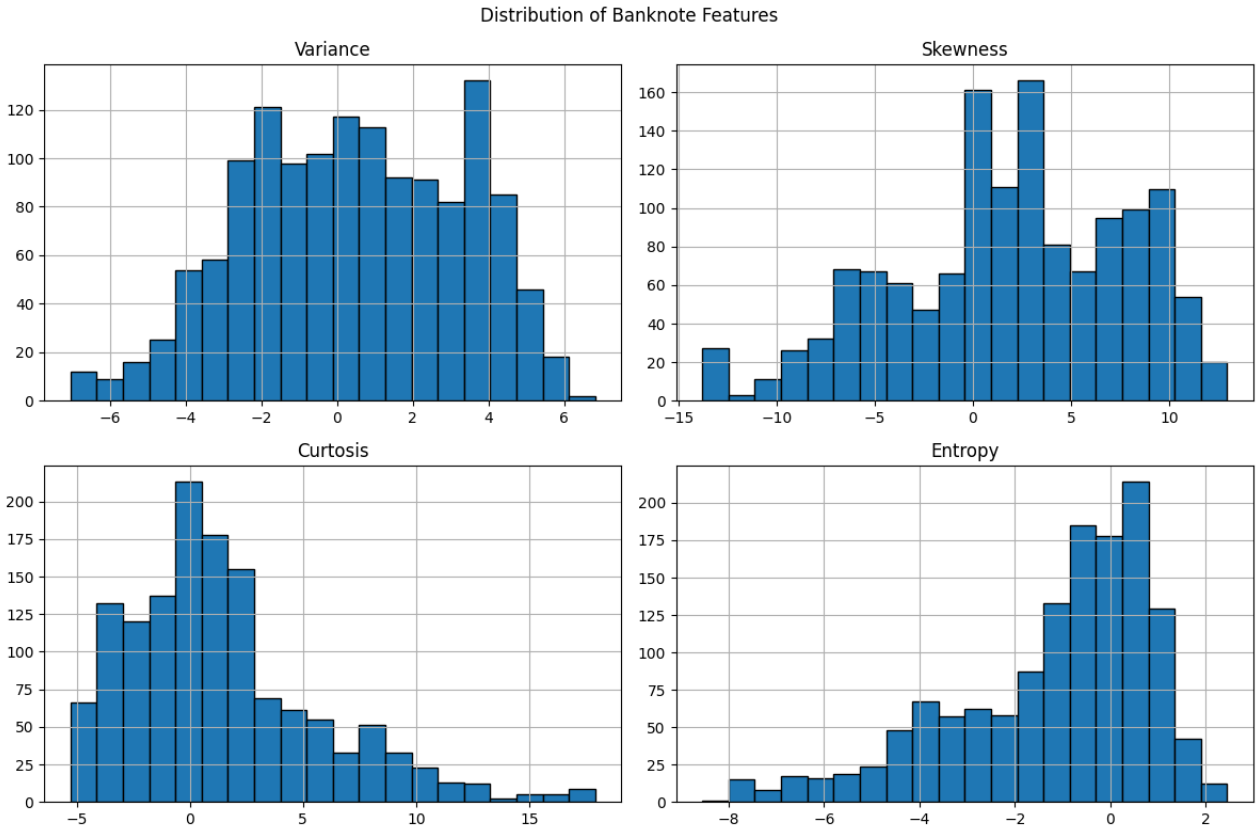


Figure 2. Histograms of the four input features

2. Feature Visualisation and Methodology

The feature histograms show that the four measurements have different distributions and numerical ranges. The scatter plots provide a visual comparison of how Class 0 and Class 1 records occupy feature space. The plotted classes show areas of separation as well as overlap, supporting the use of a multivariate classifier.

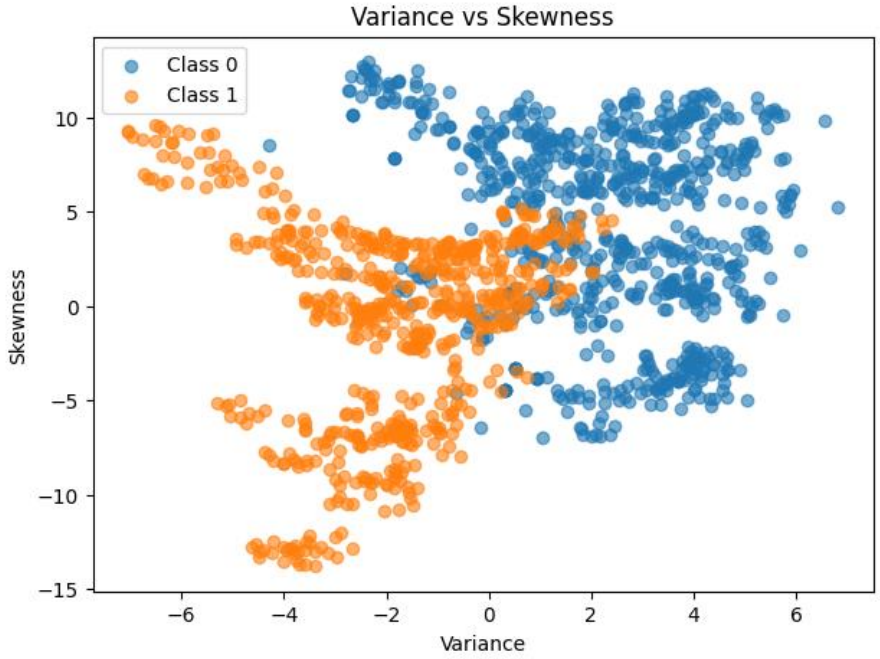


Figure 3. Variance versus Skewness by class

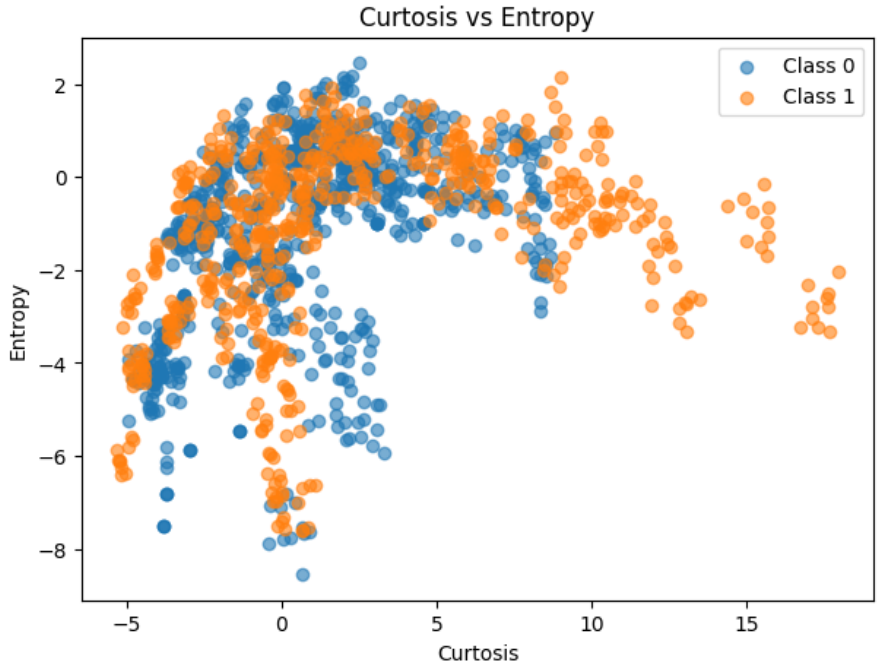


Figure 4. Curtosis versus Entropy by class

Preprocessing and Model Training

Step	Implementation
Feature/target separation	X = Variance, Skewness, Curtosis, Entropy; y = Class
Train-test split	80:20; random_state = 42; stratified by Class
Split sizes	1,097 training records and 275 test records
Feature scaling	StandardScaler fitted only on training features; then applied to test features
Classifier	LogisticRegression(max_iter=1000), trained on scaled training data
Predictions	Predicted class labels and class probabilities generated for the test set

The scaler was fitted on the training data only to prevent information from the test set influencing preprocessing. Stratification preserves the class proportions in both subsets.

3. Model Evaluation

The trained Logistic Regression model was evaluated on the 275-record test set. Class 1 was treated as the positive class when calculating precision, recall, and F1-score.

Metric	Test result
Accuracy	97.09%
Precision (Class 1)	93.85%
Recall (Class 1)	100.00%
F1-score (Class 1)	96.83%

The classification report gives Class 0 precision 1.00, recall 0.95, and F1-score 0.97; Class 1 precision 0.94, recall 1.00, and F1-score 0.97. Overall accuracy is 0.97.



Figure 5. Confusion matrix on the test set

Confusion-matrix term	Count	Interpretation
TN	145	Class 0 correctly predicted as Class 0
FP	8	Class 0 incorrectly predicted as Class 1
FN	0	Class 1 incorrectly predicted as Class 0
TP	122	Class 1 correctly predicted as Class 1

The model correctly classified 267 of 275 test records. It made eight false-positive predictions and no false-negative predictions at the default threshold.

4. Threshold Investigation

The model's Class 1 probabilities were converted to labels using thresholds 0.30, 0.50, and 0.70. A lower threshold classifies more records as Class 1; a higher threshold requires stronger predicted probability before assigning Class 1.

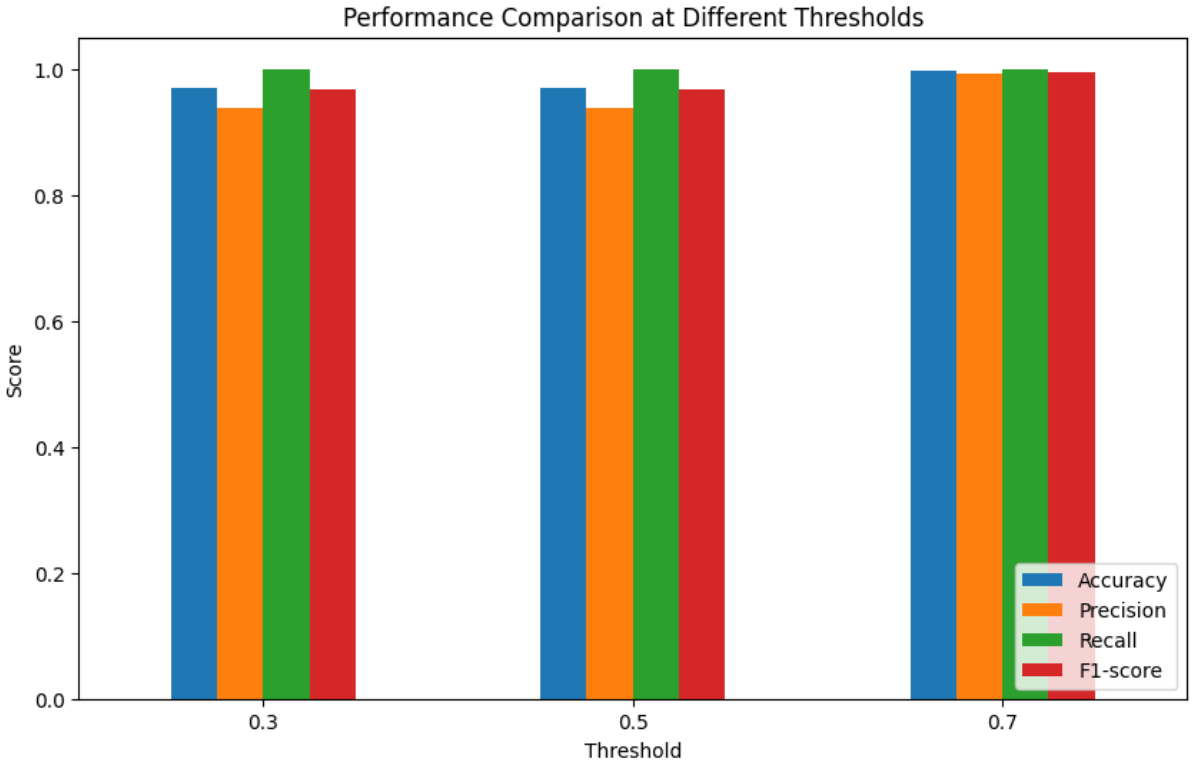


Figure 6. Performance comparison at thresholds 0.30, 0.50, and 0.70

Threshold	Accuracy	Precision	Recall	F1-score
0.30	≈0.97	≈0.94	1.00	≈0.97
0.50	0.9709	0.9385	1.0000	0.9683
0.70	≈1.00	≈0.99	1.00	≈1.00

The plotted comparison indicates that recall remains at 1.00 across the three thresholds, while precision and accuracy improve at the higher threshold. Threshold selection should consider the cost of false positives and false negatives rather than accuracy alone. The notebook's threshold plot is included as the original output.

Conclusion

This project developed a Logistic Regression classifier for the UCI Banknote Authentication dataset using Variance, Skewness, Curtosis, and Entropy as input features. The dataset contained 1,372 records and was divided into training and testing subsets using an 80:20 stratified split. StandardScaler was fitted only on the training data to avoid data leakage. On the 275-record test set, the model achieved 97.09% accuracy, 93.85% precision for Class 1, 100% recall for Class 1, and an F1-score of 96.83%. The confusion matrix contained 145 true negatives, 8 false positives, 0 false negatives, and 122 true positives. Because failing to identify a forged banknote is especially costly, recall for the forged-note class is an important metric to prioritise. The threshold investigation showed that changing the decision threshold affects positive predictions and the precision–recall balance; therefore, threshold choice should reflect the cost of different errors rather than accuracy alone. The results demonstrate strong performance on this test split, but they do not establish real-world reliability. The dataset is limited to a particular public collection of extracted features, and further validation on diverse banknotes and operating conditions would be necessary before any practical use.